

₁ Mechanistic Explainability in Reinforcement
₂ Learning

₃ Maxime Alaarabiou

₄ Saturday 26th September, 2026

Chapter 1

Conclusion

This final chapter summarizes the contributions of this manuscript with respect to the research questions. It then discusses the main limitations of this work. Finally, it presents possible directions for future research.

1.1 Summary of Contributions

This section revisits the four research questions introduced in Section ?? . For each of them, it summarizes the answer provided by this manuscript.

Research Question ?? concerned the approaches proposed in the literature to analyze the internal workings of Neural Networks (NNs). To answer it, we first made explicit the working hypothesis shared by most of this literature, the *Concept Hypothesis* (Hyp. ??). According to this hypothesis, NNs generalize by organizing their Latent Space (LS) around concepts. We then organized existing methods into four families: probing, interpretable directions, Sparse Autoencoders (SAEs), and clustering. Each family studies the LS from a different angle, but they share the goal of understanding how Latent Representations (LRs) are structured.

Research Question ?? concerned the organization of existing eXplainable Reinforcement Learning (XRL) methods [Alaarabiou et al., 2024, Alaarabiou et al., 2026b]. To answer it, we first clarified what an explanation is by placing the observer at the center of the explanatory process. Building on this frame, we proposed a taxonomy that accounts for the specificities of Reinforcement Learning (RL) and used it to review the state of the art. This review led to three main observations. First, none of the reviewed methods explains the learning process itself. Second, there are no unified metrics for evaluating explanations. Third, the field remains dominated by substitutive approaches, which replace the NN with a simpler model.

Research Question ?? concerned the use of LRs to identify the mechanisms underlying the decisions of RL agents. To answer it, we introduced Clustering Agent Representation (CAR), an XRL method based on the principles of Mechanistic Explainability (ME) [Alaarabiou et al., 2026a]. CAR trains several surrogate policies independently to reproduce the same agent. It then clusters their LRs and compares the resulting partitions. This procedure relies on the *Representation Convergence Hypothesis* (Hyp. ??), according to which indepen-

dently trained networks develop similar concepts for a given task. Applied to six Atari environments, CAR recovers similar clustering structures across surrogate policies. The qualitative analysis shows that these clusters capture interpretable aspects of the task, such as episode progression, specific game situations, and behavioral patterns.

Research Question ?? concerned the objective quantification of explanation quality. To answer it, we introduced three metrics. The *Clustering Universality* (CU) measures the consistency of the clusters across independently trained surrogate policies. The *Abstraction Score* (AS) measures how far these clusters depart from the structure of the observations, of the actions, and of an untrained network. The experiments show that these two properties evolve in opposite directions across the network. The earliest layers are the most consistent, but also the least abstract. The *Adjusted Clustering Universality* (ACU) combines both properties into a single measure and provides a practical criterion for selecting the layer to analyze. These metrics make it possible to compare competing hypotheses about the structure of a LS without relying on manual inspection.

Taken together, these contributions form a coherent chain. The *Mechanistic Hypothesis* (Hyp. ??) states that NNs can be explained through their internal mechanisms. The *Concept Hypothesis* (Hyp. ??) states that these mechanisms are organized around concepts. The *Representation Convergence Hypothesis* (Hyp. ??) states that these concepts reflect the task rather than a particular network. CAR translates these hypotheses into an empirical procedure, and the experiments provide support for them.

1.2 Limitations

Although the research presented in this thesis has produced conclusive results, it remains subject to limitations. This section discusses the five main limitations identified at this stage.

CAR describes how the LS of an agent is organized, but not what each cluster means. The metrics quantify the consistency and abstraction of the clusters, but they say nothing about their content. In this manuscript, the meaning of each cluster was assigned through manual inspection of the corresponding observations. This step is time-consuming, subjective, and difficult to scale to a large number of clusters or environments.

The clusters identified by CAR are not yet directly related to the decisions of the agent. We observe that the agent distinguishes certain situations, but not how these distinctions lead to a given action. As a result, CAR cannot yet explain a specific failure or reveal a bias in the behavior of the agent.

The experiments were conducted on six Atari environments with relatively small networks. These environments are diverse, but they remain limited in size and complexity. Whether CAR scales to more complex environments and larger networks has not yet been established. In addition, only one clustering method was evaluated, based on k -means with Euclidean distance. Other distances and other clustering methods could reveal different structures in the LS. Since the pipeline of CAR is modular, these alternatives can be tested and compared using the proposed metrics.

The proposed metrics have only been applied to clustering. The other families of methods reviewed in Chapter ?? have not yet been evaluated within this

framework. This is particularly the case for methods based on the *Superposition Hypothesis* (Hyp. ??), such as SAEs. These methods are widely used, but their assumptions remain debated. Our metrics could be used to test these assumptions empirically and quantitatively, and to compare these methods with CAR on the same tasks.

1.3 Perspectives

The limitations discussed above open several directions for future research. This section presents three of them, from the shortest to the longest term.

A first direction is to automate the semantic interpretation of the clusters. One approach is to rely on vision-language models or Large Language Models (LLMs). Given observations from the same cluster, such models could generate a natural language description of what distinguishes it from the others. This approach would provide a textual description for each cluster, helping identify its specific semantics and fully automate the CAR method.

A second direction is to relate the identified clusters to the strategy followed by the agent. The first step was to verify whether the latent space could be used to identify meaningful concepts. The results obtained with CAR show that this is a promising direction. The next step is to use these concepts to explain the agent's strategy. One possible approach would be to group similar states from the agent's perspective, thereby providing a simplified representation of the environment and capturing its relevant dynamics.

A third direction concerns the evaluation of explanations. Structural metrics are necessary, but they are not sufficient. The value of an explanation also depends on its effect on the observer. Measuring this effect requires experimental protocols from disciplines such as cognitive science and psychology. This would help determine whether the ultimate goal of explainability has been achieved.

112 Acronyms

- 113 **ACU** *Adjusted Clustering Universality* 2
- 114 **AS** *Abstraction Score* 2

- 115 **CAR** Clustering Agent Representation 2, 3
- 116 **CU** *Clustering Universality* 2

- 117 **LLM** Large Language Model 3
- 118 **LR** Latent Representation 1, 2
- 119 **LS** Latent Space 1–3

- 120 **ME** Mechanistic Explainability 1–3

- 121 **NN** Neural Network 1–3

- 122 **RL** Reinforcement Learning 1, 2

- 123 **SAE** Sparse Autoencoder 1, 3

- 124 **XRL** eXplainable Reinforcement Learning 1–3

Bibliography

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